TASK SPECIFICATION: P9-01 PHASE P9 — PRODUCTION DEPLOYMENT, CI/CD & OPERATIONS

Production Orchestration & Secrets Management

Git Branch: `chore/P9-01-prod-config` · Target Pull Request: chore/P9-01-prod-config

Core Objective & Technical Scope:
Hardened production Docker Compose configuration, TLS termination, reverse proxy rules, and externalized secrets injection.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`chore/P9-01-prod-config`
Upstream Dependencies	P8-05		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author hardened docker-compose.prod.yml with non-root security contexts.
2. Configure Nginx reverse proxy with TLS certificate termination and HTTP/2.
3. Externalize all production secrets via environment variables.
4. Document complete production deployment runbook.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `docker-compose.prod.yml`
- `docs/runbooks/deployment.md`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Zero secrets or sensitive defaults stored in git repository.
- Production stack starts cleanly with all services healthy.

VERIFICATION & TEST COVERAGE STRATEGY

- Production compose startup drill
- TLS certificate configuration test